

MSELoss#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.loss.MSELoss.html>

Description:

Creates a criterion that measures the mean squared error (squared L2 norm) between each element in the input `xxx` and target `yyy`. The unreduced (i.e. with reduction set to 'none') loss can be described as: where `NNN` is the batch size. If reduction is not 'none' (default 'mean'), then: `xxx` and `yyy` are tensors of arbitrary shapes with a total of `NNN` elements each. The mean operation still operates over all the elements, and divides by `NNN`.

Function Signature

```
class torch.nn.modules.loss.MSELoss(size_average=None,
reduce=None, reduction='mean')[source]#
```

Parameters

Shape:

```
forward(input, target)[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> loss = nn.MSELoss()
>>> input = torch.randn(3, 5, requires_grad=True)
>>> target = torch.randn(3, 5)
>>> output = loss(input, target)
>>> output.backward()
```

Detailed Documentation

Documentation

Creates a criterion that measures the mean squared error (squared L2 norm) between each element in the input `xxx` and target `yyy`.

The unreduced (i.e. with reduction set to 'none') loss can be described as:

where `NNN` is the batch size. If reduction is not 'none' (default 'mean'), then:

`xxx` and `yyy` are tensors of arbitrary shapes with a total of `NNN` elements each.

The mean operation still operates over all the elements, and divides by `NNN`.

The division by NNN can be avoided if one sets `reduction = 'sum'`.

`size_average` (bool, optional) – Deprecated (see `reduction`). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field `size_average` is set to `False`, the losses are instead summed for each minibatch. Ignored when `reduce` is `False`. Default: `True`

`reduce` (bool, optional) – Deprecated (see `reduction`). By default, the losses are averaged or summed over observations for each minibatch depending on `size_average`. When `reduce` is `False`, returns a loss per batch element instead and ignores `size_average`. Default: `True`

`reduction` (str, optional) – Specifies the reduction to apply to the output: `'none'` | `'mean'` | `'sum'`. `'none'`: no reduction will be applied, `'mean'`: the sum of the output will be divided by the number of elements in the output, `'sum'`: the output will be summed. Note: `size_average` and `reduce` are in the process of being deprecated, and in the meantime, specifying either of those two args will override `reduction`. Default: `'mean'`

Input: $(*)^{(*)} (*)$, where $***$ means any number of dimensions.

Target: $(*)^{(*)} (*)$, same shape as the input.

Runs the forward pass.